



SQL for Data Science

This course is part of [Learn SQL Basics for Data Science Specialization](#)

Instructor: [Sadie St. Lawrence](#)

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Already enrolled

724,736 already enrolled

Included with **coursera** PLUS

4 modules

Gain insight into a topic and learn the fundamentals.

4.6 ★

17 208 reviews

Beginner level

No prior experience required

Flexible schedule

2 weeks at 10 hours a week
Learn at your own pace

👍 90%

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- About
- Outcomes
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What you'll learn

- ✓ Identify a subset of data needed from a column or set of columns and write a SQL query to limit to those results.
- ✓ Use SQL commands to filter, sort, and summarize data.
- ✓ Create an analysis table from multiple queries using the UNION operator.
- ✓ Manipulate strings, dates, & numeric data using functions to integrate data from different sources into fields with the correct format for analysis.

Skills you'll gain

- Data Wrangling
- Descriptive Analytics
- Database Management
- Data Manipulation
- Data Governance
- Data Modeling
- Data Transformation
- Data Analysis
- Data Literacy
- Data Quality
- SQL
- Data Science

Tools you'll learn

- Relational Databases
- Query Languages

Details to know

in

Shareable certificate

Add to your LinkedIn profile

📅

Assessments

15 assignments¹
[AI Graded see disclaimer](#)

🗣️

Taught in English

25 languages available

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Build your subject-matter expertise

This course is part of the [Learn SQL Basics for Data Science Specialization](#)
When you enroll in this course, you'll also be enrolled in this Specialization.

- Learn new concepts from industry experts
- Gain a foundational understanding of a subject or tool
- Develop job-relevant skills with hands-on projects
- Earn a shareable career certificate



There are 4 modules in this course

As data collection has increased exponentially, so has the need for people skilled at using and interacting with data; to be able to think critically, and provide insights to make better decisions and optimize their businesses. This is a data scientist, “part mathematician, part computer scientist, and part trend spotter” (SAS Institute, Inc.). According to Glassdoor, being a data scientist is the best job in America; with a median base salary of \$110,000 and thousands of job openings at a time. The skills necessary to be a good data scientist include being able to retrieve and work with data, and to do that you need to be well versed in SQL, the standard language for communicating with database systems.

This course is designed to give you a primer in the fundamentals of SQL and working with data so that you can begin analyzing it for data science purposes. You will begin to ask the right questions and come up with good answers to deliver valuable insights for your organization. This course starts with the basics and assumes you do not have any knowledge or skills in SQL. It will build on that foundation and gradually have you write both simple and complex queries to help you select data from tables. You'll start to work with different types of data like strings and numbers and discuss methods to filter and pare down your results.

You will create new tables and be able to move data into them. You will learn common operators and how to combine the data. You will use case statements and concepts like data governance and profiling. You will discuss topics on data, and practice using real-world programming assignments. You will interpret the structure, meaning, and relationships in source data and use SQL as a professional to shape your data for targeted analysis purposes.

Although we do not have any specific prerequisites or software requirements to take this course, a simple text editor is recommended for the final project. So what are you waiting for? This is your first step in landing a job in the best occupation in the US and soon the world!

[Read more](#)

Getting Started and Selecting & Retrieving Data with SQL

Module 1 • 4 hours to complete

[Module details](#)

In this module, you will be able to define SQL and discuss how SQL differs from other computer languages. Additionally, compare and contrast the roles of a database administrator and a data scientist, and explain the differences between one-to-one, one-to-many, and many-to-many relationships with databases. You will also be able to use the SELECT statement and talk about some basic syntax rules, add comments in your code, and synthesize their importance.

What's included

- 📺 10 videos
- 📖 3 readings
- 📅 5 assignments
- 💬 1 discussion prompt

Hide info about module content

📺 10 videos • Total 51 minutes

Course Introduction • 4 minutes

What is SQL Anyway? • 7 minutes

Data Models, Part 1: Thinking About Your Data • 6 minutes

Data Models, Part 2: The Evolution of Data Models • 4 minutes

Data Models, Part 3: Relational vs. Transactional Models • 6 minutes

Instructor

Instructor ratings 4.6 ★ (4842 ratings)



Sadie St. Lawrence
University of California, Davis
4 Courses • 745,312 learners

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University of California, Davis
[Learn more](#)

- Retrieving Data with a SELECT Statement • 5 minutes
- Creating Tables • 8 minutes
- Creating Temporary Tables • 4 minutes
- Adding Comments to SQL • 5 minutes
- Summary • 2 minutes

 3 readings • Total 65 minutes

Course Navigation and Online Safety Statement • 15 minutes

SQL Overview • 15 minutes

Data Modeling and ER Diagrams • 35 minutes

 5 assignments • Total 120 minutes

Module 1 Quiz • 30 minutes

Module 1 Coding Questions • 10 minutes

Introduction to SQL for Data Science • 20 minutes

Let's Practice! • 30 minutes

Practice Simple Select Queries • 30 minutes

 1 discussion prompt • Total 30 minutes

Comparing NoSQL and SQL • 30 minutes

Filtering, Sorting, and Calculating Data with SQL

[Module details](#) ^

Module 2 • 3 hours to complete

In this module, you will be able to use several more new clauses and operators including WHERE, BETWEEN, IN, OR, NOT, LIKE, ORDER BY, and GROUP BY. You will be able to use the wildcard function to search for more specific or parts of records, including their advantages and disadvantages, and how best to use them. You will be able to discuss how to use basic math operators, as well as aggregate functions like AVERAGE, COUNT, MAX, MIN, and others to begin analyzing our data.

What's included

 9 videos  1 reading  3 assignments

Hide info about module content ^

 9 videos • Total 49 minutes

Module Introduction • 2 minutes

Basics of Filtering with SQL • 9 minutes

Advanced Filtering: IN, OR, and NOT • 7 minutes

Using Wildcards in SQL • 7 minutes

Sorting with ORDER BY • 5 minutes

Math Operations • 4 minutes

Aggregate Functions • 7 minutes

Grouping Data with SQL • 6 minutes

Putting it All Together • 3 minutes

 1 reading • Total 25 minutes

SQL for Various Data Science Languages • 25 minutes

 3 assignments • Total 105 minutes

Module 2 Quiz • 30 minutes

Module 2 Coding Assignment • 45 minutes

Module 2 Practice Quiz • 30 minutes

Subqueries and Joins in SQL

[Module details](#) ^

Module 3 • 3 hours to complete

In this module, you will be able to discuss subqueries, including their advantages and disadvantages, and when to use them. You will be able to recall the concept of a key field and discuss how these help us link data together with JOINS. You will be able to identify and define several types of JOINS, including the Cartesian join, an inner join, left and right joins, full outer joins, and a self join. You will be able to use aliases and pre-qualifiers to make your SQL code cleaner and efficient.

What's included

 10 videos  2 readings  3 assignments  1 discussion prompt

Hide info about module content ^

 10 videos • Total 58 minutes

Module Introduction • 1 minute

Using Subqueries • 7 minutes

Subquery Best Practices and Considerations • 8 minutes

Joining Tables: An Introduction • 5 minutes

Cartesian (Cross) Joins • 4 minutes

Inner Joins • 8 minutes

Aliases and Self Joins • 6 minutes

Advanced Joins: Left, Right, and Full Outer Joins • 8 minutes

Unions • 5 minutes

Summary • 6 minutes

 2 readings • Total 20 minutes

SQL and Python • 10 minutes

Union and Union All • 10 minutes

 3 assignments • Total 105 minutes

Module 3 Quiz • 30 minutes

Module 3 Coding Assignment • 45 minutes

Practice Quiz - Writing Queries • 30 minutes

 1 discussion prompt • Total 20 minutes

What do you think you'll use? • 20 minutes

Modifying and Analyzing Data with SQL

[Module details](#) ^

Module 4 • 5 hours to complete

In this module, you will be able to discuss how to modify strings by concatenating, trimming, changing the case, and using the substring function. You will be able to discuss the date and time strings specifically. You will be able to use case statements and finish this module by discussing data governance and profiling. You will also be able to apply fundamental principles when using SQL for data science. You'll be able to use tips and tricks to apply SQL in a data science context.

What's included

11 videos

3 readings

4 assignments

1 discussion prompt

Hide info about module content ^

11 videos • Total 56 minutes

Module Introduction • 1 minute

Working with Text Strings • 7 minutes

Working with Date and Time Strings • 5 minutes

Date and Time Strings Examples • 5 minutes

Case Statements • 7 minutes

Views • 7 minutes

Data Governance and Profiling • 6 minutes

Using SQL for Data Science, Part 1 • 6 minutes

Using SQL for Data Science, Part 2 • 6 minutes

USDA Acknowledgment Video • 2 minutes

Course Summary • 2 minutes

3 readings • Total 61 minutes

Additional SQL Resources to Explore • 15 minutes

Final Project Instructions, Scripts, & Datasets • 45 minutes

USDA Acknowledgment • 1 minute

4 assignments • Total 155 minutes

Module 4 Quiz • 30 minutes

Module 4 Coding Questions • 35 minutes

Final Project • 45 minutes

Practice Final Project • 45 minutes

1 discussion prompt • Total 15 minutes

How do you Plan on Using SQL in the Future? • 15 minutes

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University of California, Davis

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Course

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Learner reviews

Showing 3 of 17208

★ 4.6

17,208 reviews

5 stars

4 stars

3 stars

2 stars

1 star

73.46%

19.88%

3.57%

1.45%

1.62%

N

NG

★ 4 • Reviewed on Aug 28, 2020

The course is nice but I suppose the PowerPoint presentation that accompanies it could use some effects to show us the clauses one by one, as they are spoke. That would make understanding them easier.

R

RN

★ 5 • Reviewed on Jun 14, 2020

Better than I hoped for. I learned everything I needed in order to feel competent in the basics of SQL. It was just the right amount of instruction and challenge, perfect for a beginner like me.

D

DA

★ 5 • Reviewed on Jun 22, 2021

Thank you for the material that was presented well and clearly. I got to know better how to use SQL for data science. Hopefully, by learning this, I can implement it in the future as a Data Analyst.

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Frequently asked questions

✓ When will I have access to the lectures and assignments?

✓ What will I get if I subscribe to this Specialization?

✓ Is financial aid available?

?

More questions

[Visit the learner help center](#)

¹ Some assignments in this course are AI-graded. For these assignments, your data will be used in accordance with [Coursera's Privacy Notice](#).

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